



Derivatives of Exponential Functions

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Apply the derivative rule for e^u to differentiate natural exponential functions
- Apply the derivative rule for a^u to differentiate general exponential functions
- Use the chain rule with exponential functions involving polynomial exponents

Find the derivative of each function using the rules for differentiating exponential functions, and simplify your answer.

1. Find the derivative of the function.

$$y = e^{7x}$$

Answer: _____

2. Find the derivative of the function.

$$y = 5^{7x}$$

Answer: _____

3. Find the derivative of the function.

$$y = 5e^{2x^3 + 3x}$$

Answer: _____

4. Find the derivative of the function.

$$y = e^{4x^2}$$

Answer: _____

5. Find the derivative of the function.

$$y = 3^{x^2 + 1}$$

Answer: _____

6. Find the derivative of the function.

$$f(x) = e^{\sqrt{x}}$$

Answer: _____

7. Find the derivative of the function.

$$y = 2^{5x-1}$$

Answer: _____



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8. Find the derivative of the function.

$$y = 4e^{x^3 - 2x}$$

Answer: _____

9. Find the derivative of the function.

$$y = 10^{3x^2}$$

Answer: _____

10. Find the derivative of the function.

$$f(x) = e^{(3x)^5}$$

Answer: _____





Remind students that the derivative of e^u is e^u times du/dx , and the derivative of a^u is a^u times $\ln(a)$ times du/dx .

Solutions

1. Find the derivative of the function.

$$y = e^{7x}$$

→ Identify u equals $7x$, so du/dx equals 7 .

→ Apply the rule: derivative of e to the u equals e to the u times du/dx .

→ Multiply to get 7 times e to the $7x$.

Answer: $y' = 7e^{7x}$

2. Find the derivative of the function.

$$y = 5^{7x}$$

→ Identify the base a equals 5 and exponent u equals $7x$.

→ Apply the rule: derivative of a to the u equals a to the u times $\ln(a)$ times du/dx .

→ Since du/dx equals 7 , multiply to get 7 times 5 to the $7x$ times $\ln 5$.

Answer: $y' = 7 \cdot 5^{7x} \ln 5$

3. Find the derivative of the function.

$$y = 5e^{2x^3 + 3x}$$

→ Keep the constant 5 outside and focus on differentiating e to the $2x$ cubed plus $3x$.

→ Apply the rule: derivative equals e to the u times du/dx where u equals $2x$ cubed plus $3x$.

→ Compute du/dx equals $6x$ squared plus 3 .

→ Multiply by the constant 5 using the distributive property to get $30x$ squared plus 15 times e to the $2x$ cubed plus $3x$.

Answer: $y' = (30x^2 + 15)e^{2x^3 + 3x}$

4. Find the derivative of the function.

$$y = e^{4x^2}$$

→ Let u equal $4x$ squared so du/dx equals $8x$.

→ Apply the e to the u rule: e to the u times du/dx .

→ Write the final answer as $8x$ times e to the $4x$ squared.

Answer: $y' = 8x e^{4x^2}$

5. Find the derivative of the function.

$$y = 3^{x^2 + 1}$$

→ Identify base a equals 3 and exponent u equals x squared plus 1 .

→ Apply the rule: a to the u times $\ln(a)$ times du/dx .

→ Since du/dx equals $2x$, the derivative is $2x$ times 3 to the x squared plus 1 times $\ln 3$.

Answer: $y' = 2x \cdot 3^{x^2 + 1} \ln 3$



6. Find the derivative of the function.

$$f(x) = e^{\sqrt{x}}$$

→ Let u equal the square root of x , which is x to the one-half power.

→ Compute du/dx equals one over two times the square root of x .

→ Apply the e to the u rule and multiply to get e to the square root of x divided by 2 times the square root of x .

Answer: $f'(x) = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$

7. Find the derivative of the function.

$$y = 2^{5x-1}$$

→ Identify base a equals 2 and exponent u equals $5x$ minus 1.

→ Apply the rule: a to the u times $\ln(a)$ times du/dx .

→ Since du/dx equals 5, the answer is 5 times 2 to the $5x$ minus 1 times $\ln 2$.

Answer: $y' = 5 \cdot 2^{5x-1} \ln 2$

8. Find the derivative of the function.

$$y = 4e^{x^3-2x}$$

→ Keep the constant 4 outside and focus on e to the x cubed minus $2x$.

→ Let u equal x cubed minus $2x$ so du/dx equals $3x$ squared minus 2.

→ Apply the e to the u rule and multiply by the constant 4 to get 4 times the quantity $3x$ squared minus 2 times e to the x cubed minus $2x$.

Answer: $y' = 4(3x^2 - 2)e^{x^3-2x}$

9. Find the derivative of the function.

$$y = 10^{3x^2}$$

→ Identify base a equals 10 and exponent u equals $3x$ squared.

→ Apply the rule: a to the u times $\ln(a)$ times du/dx .

→ Compute du/dx equals $6x$ and multiply to get $6x$ times 10 to the $3x$ squared times $\ln 10$.

Answer: $y' = 6x \cdot 10^{3x^2} \ln 10$

10. Find the derivative of the function.

$$f(x) = e^{(3x)^5}$$

→ Simplify the exponent: $3x$ raised to the 5th equals 243 times x to the 5th.

→ Let u equal 243 x to the 5th so du/dx equals 1215 x to the 4th.

→ Apply the e to the u rule to get 1215 x to the 4th times e to the 243 x to the 5th.

Answer: $f'(x) = 1215x^4 e^{243x^5}$

